

Weight decomposition for semi-simple Lie algebra

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Problem 1. Let V be the vector space of homogeneous polynomials of degree 3 in variables x, y, z , equipped with the natural action of $\mathfrak{sl}_3(\mathbb{C})$.

1. Let

$$\mathfrak{h} = \{\text{diag}(h_1, h_2, h_3) \mid h + 1h_2 + h_3 = 0\} \subseteq \mathfrak{sl}_3(\mathbb{C}).$$

Write down the root space decomposition of $\mathfrak{sl}_3(\mathbb{C})$ with respect to $\mathfrak{h}$.

2. For each simple root, write down the corresponding $\mathfrak{sl}_2$ -triples explicitly as 3×3 matrices.

3. Write down explicitly the action of $\mathfrak{sl}_3(\mathbb{C}) \subseteq \mathfrak{gl}_3(\mathbb{C})$ on V .

4. Compute the weight decomposition of V with respect to $\mathfrak{h}$. List all weights and their multiplicities.

5. Draw the weight lattice for this representation, indicating all weights and their multiplicities.

6. Choose one of the $\mathfrak{sl}_2$ -triples and compute explicitly how it acts on the monomial basis $x^a y^b z^c$. Describe the resulting $\mathfrak{sl}_2$ -strings in V .

7. Describe the Weyl group of $\mathfrak{sl}_3(\mathbb{C})$ and compute its action on the weight lattice. Verify that the set of weights is invariant under this action and determine the Weyl group orbits.

Proof. (1) Let $h = \text{diag}(h_1, h_2, h_3) \in \mathfrak{h}$. The linear functionals $\mathfrak{h}^*$ are generated by $L_i h = h_i$, $1 \leq i \leq 3$. Let $E_{ij} = \delta_{ij}^j$ denote the standard basis matrices of $\mathfrak{gl}_3(\mathbb{C})$. Then note that $[h, E_{ij}] = (h_i - h_j)E_{ij}$. Rewrite this as

$$[h, E_{ij}] = (L_i h - L_j h)E_{ij} = (L_i - L_j)(h)E_{ij}.$$

Thus we find that the roots are generated by $L_i - L_j$, and that the root space $\mathfrak{g}_{L_i - L_j}$ is $\text{span}(E_{ij})$ as a $\mathbb{C}$ -vector space. The root space decomposition is

$$\mathfrak{sl}_3(\mathbb{C}) = \mathfrak{h} \oplus \bigoplus_{1 \leq i \neq j \leq 3} \mathfrak{g}_{L_i - L_j}.$$

(2) Write $\alpha_1 = L_1 - L_2$, $\alpha_2 = L_2 - L_3$, and $\alpha_3 = L_1 - L_3$. We claim the simple roots are $L_1 - L_2$ and $L_2 - L_3$. This is immediately shown: $\{\alpha_1, \alpha_2\}$ clearly forms a basis for $\mathfrak{h}^*$, and additionally, $\alpha_3 = \alpha_1 + \alpha_2$.

We now find the $\mathfrak{sl}_2$ -triples for the simple roots. Since $\mathfrak{g}_{\alpha_1} = \text{span}(E_{12})$, we set $e_1 = cE_{12}$ for some $c \in \mathbb{C}^\times$. Similarly, $\mathfrak{g}_{-\alpha_1} = \text{span}(E_{21})$ implies $f_1 = dE_{21}$ for some $d \in \mathbb{C}^\times$. We use the $\mathfrak{sl}_2$ commutator relations to compute

$$h_1 = [e_1, f_1] = cd(E_{12}E_{21} - E_{21}E_{12}) = cd(E_{11} - E_{22}).$$

Note that $h_1 \in \mathfrak{h}$ implies $\alpha_1(h_1)e_1 = [h_1, cE_{12}] = 2e_1$, and thus $\alpha_1(h_1) = 2$. It follows that

$$2 = \alpha_1(h_1) = cd(L_1(E_{11}) - L_2(E_{22})) = 2cd.$$

We set $c = d = 1$. This yields

$$e_1 = E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad f_1 = E_{21} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad h_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The same logic shows that $e_2 = E_{23}$, $f_2 = E_{32}$, and $h_2 = E_{22} - E_{33}$. Thus

$$e_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad f_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad h_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(3) We start with the obvious representation on $\mathbb{C}^3 = \text{span}\{x_1, x_2, x_3\}$ given by matrix multiplication $E_{ij}x_k = \delta_{jk}x_i$. Then we note that V is the third symmetric power $V \cong \text{Sym}^3(\mathbb{C}^3)$, and we can extend Lie algebra actions to symmetric powers by acting as derivations:

$$X(pq) = X(p)q + pX(q), \quad X \in \mathfrak{sl}_3(\mathbb{C}), \quad p, q \in \mathbb{C}^3.$$

The Leibniz rule implies that $\mathfrak{sl}_3(\mathbb{C})$ should act via differential operators. We start by noting that $E_{ij} \cdot x_k = \delta_{jk}x_i = x_i \partial_j x_k$. Since derivatives satisfy the Leibniz rule, this extends naturally to the symmetric power V . Thus we can define the action of E_{ij} on V by

$$E_{ij}p = x_i \frac{\partial p}{\partial x_j}, \quad p \in V.$$

We then of an arbitrary $X \in \mathfrak{sl}_3(\mathbb{C})$ on $p \in V$ by noting that $\{E_{ij}\}_{1 \leq i, j \leq 3}$ is a basis for $\mathfrak{sl}_3(\mathbb{C})$.

(4) The space V has a basis $B = \{x^a y^b z^c \mid a + b + c = 3, \quad a, b, c \geq 0\}$ of degree 3 monomials. Let $h = \text{diag}(h_1, h_2, h_3) \in \mathfrak{h}$. Then by (3),

$$h \cdot x^a y^b z^c = \left(h_1 x \frac{\partial}{\partial x} + h_2 y \frac{\partial}{\partial y} + h_3 z \frac{\partial}{\partial z} \right) (x^a y^b z^c) = (ah_1 + bh_2 + ch_3) x^a y^b z^c.$$

This corresponds to a weight $\lambda_{a,b,c} = aL_1 + bL_2 + cL_3$, since $\lambda_{a,b,c}(h)x^a y^b z^c = h \cdot x^a y^b z^c$. Each tuple (a, b, c) of nonnegative integers thus determines a unique weight, meaning all weights have a multiplicity of 1.

(5) See [?], which is on the next page due to LaTeX figure placement being trash. Warning: I cannot draw well.

(6) Consider the α_1 $\mathfrak{sl}_2$ -triple (e_1, f_1, h_1) . As polynomial differential operators these are

$$e_1 = x \frac{\partial}{\partial y}, \quad f_1 = y \frac{\partial}{\partial x}, \quad h_1 = x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}.$$

This yields

$$e_1 \cdot x^a y^b z^c = b x^{a+1} y^{b-1} z^c, \quad f_1 \cdot x^a y^b z^c = a x^{a-1} y^{b+1} z^c, \\ h_1 \cdot x^a y^b z^c = (a - b) x^a y^b z^c.$$

I assume $\mathfrak{sl}_2$ -strings are given by applying the raising and lowering operators, so we compute those for α_1 . Since this $\mathfrak{sl}_2$ -action leaves the z^c component invariant, there are four strings as follows.

- ($c = 0$) There is a 4-dimensional string

$$\langle y^3 \rangle \xleftarrow[f_1]{e_1} \langle xy^2 \rangle \xleftarrow[f_1]{e_1} \langle x^2 y \rangle \xleftarrow[f_1]{e_1} \langle x^3 \rangle.$$

This string has highest weight 3.

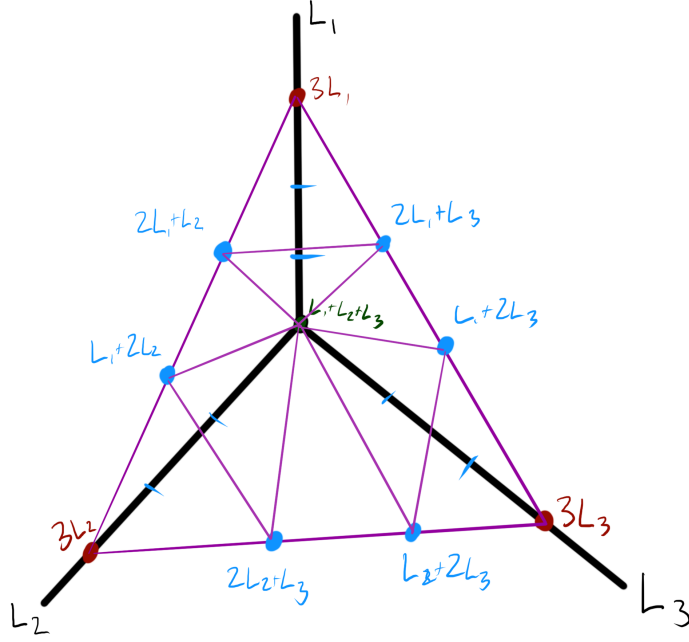


Figure 1: Weight lattice for $\mathfrak{sl}_3(\mathbb{C})$

- ($c = 1$) There is a 3-dimensional string

$$\langle x^2 z \rangle \xrightleftharpoons[f_1]{e_1} \langle xyz \rangle \xrightleftharpoons[f_1]{e_1} \langle y^2 z \rangle.$$

This string has highest weight 2.

- ($c = 2$) There is a 2-dimensional string

$$\langle xz^2 \rangle \xrightleftharpoons[f_1]{e_1} \langle yz^2 \rangle.$$

This string has highest weight 1.

- ($c = 3$) Since z^3 is annihilated by e_1 and f_1 , we have that $\langle z^3 \rangle$ is a trivial 1-dimensional $\mathfrak{sl}_2$ -string of highest weight 0.

(7) We use the inner product $\langle L_i, L_j \rangle = \delta_{ij}$ to prove that the Weyl group of $\mathfrak{sl}_3(\mathbb{C})$ is isomorphic to S_3 . The Weyl group is generated by reflections

$$s_\alpha(\lambda) = \lambda - \frac{2\langle \lambda, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha,$$

where α is a simple root and $\lambda \in \mathfrak{h}^*$. First, note that

$$\langle \alpha_1, \alpha_1 \rangle = \langle L_1 - L_2, L_1 - L_2 \rangle = 2,$$

$$\langle \alpha_2, \alpha_2 \rangle = \langle L_2 - L_3, L_2 - L_3 \rangle = 2.$$

Thus $s_\alpha(\lambda) = \lambda - \langle \lambda, \alpha \rangle \alpha$. Let $\lambda = aL_1 + bL_2 + cL_3 \in \mathfrak{h}^*$. Then we compute

$$\langle \lambda, \alpha_1 \rangle = a - b, \quad \langle \lambda, \alpha_2 \rangle = b - c.$$

Thus

$$\begin{aligned} s_{\alpha_1}(\lambda) &= aL_1 + bL_2 + cL_3 - (a - b)(L_1 - L_2) = aL_1 + bL_2 + cL_3 - aL_1 + bL_1 + aL_2 - bL_2 \\ &= bL_1 + aL_2 + cL_3; \\ s_{\alpha_2}(\lambda) &= aL_1 + bL_2 + cL_3 - bL_2 + cL_2 + bL_3 - cL_3 = aL_1 + cL_2 + bL_3. \end{aligned}$$

We thus find that if we write $\lambda(a, b, c)$ as a vector in the $\{L_i\}$ -basis, then the Weyl group W is isomorphic to the symmetric group S_3 : s_{α_1} acts as the transposition (12), s_{α_2} acts as the transposition (23), and these two elements generate S_3 (note that (12)(23) = (123) so you get 3-cycles).

With this in mind, we consider the action of $W \cong S^3$ on the weight lattice Π . Recall that weights are $\Pi \ni \lambda = aL_1 + bL_2 + cL_3 = (a, b, c)$ with $a + b + c = 3$ nonnegative integers. If we permute the indicies, then they are still all nonnegative and still sum to 3, so $S_3\Pi \subseteq \Pi$. There are three distinct orbits.

- If $(a, b, c) = (3, 0, 0)$, then the orbit of λ is $\{(3, 0, 0), (0, 3, 0), (0, 0, 3)\}$. Explicitly, this is $\{3L_1, 3L_2, 3L_3\}$.
- If $(a, b, c) = (2, 1, 0)$, then the orbit of λ is all permutations of $(2, 1, 0)$:

$$\{2L_1 + L_2, 2L_1 + L_3, L_1 + 2L_2, 2L_2 + L_3, L_1 + 2L_3, L_2 + 2L_3\}.$$

- If $(a, b, c) = (1, 1, 1)$, then the orbit of λ is the only permutation of $(1, 1, 1)$: $\{L_1 + L_2 + L_3\}$.

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